

OLIST E-Commerce

Delivery and Operational Performance

By Andrea Yining Liu



Context



le wagon

Project Overview & Objectives



- Olist is an e-commerce platform selling products online in Brazil.
- This project analysed the orders from 2016 to 2018, including delivery timelines and seller performance in different states.
- By evaluating delivery speed, factors affecting delivery, and seller performance by region, customer experience and operational efficiency may be improved.



le wagon

Business needs



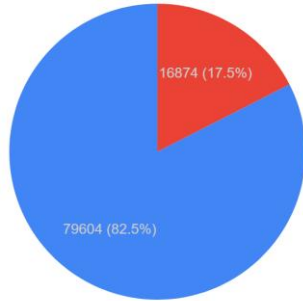
le wagon

Core Operational Risks

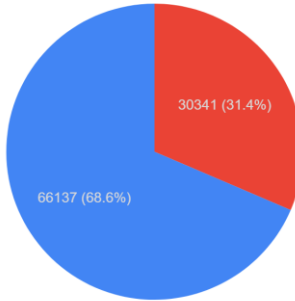
Delivery Delays
Average Delivery Days
8.88

Unpredictable delivery timelines may impact customer satisfaction rates. The delivery stage experiences the highest rate of delays.

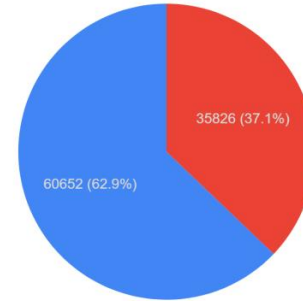
Orders Exceeding Average Approval Time



Orders Exceeding Average Dispatch Time



Orders Exceeding Average Delivery Time



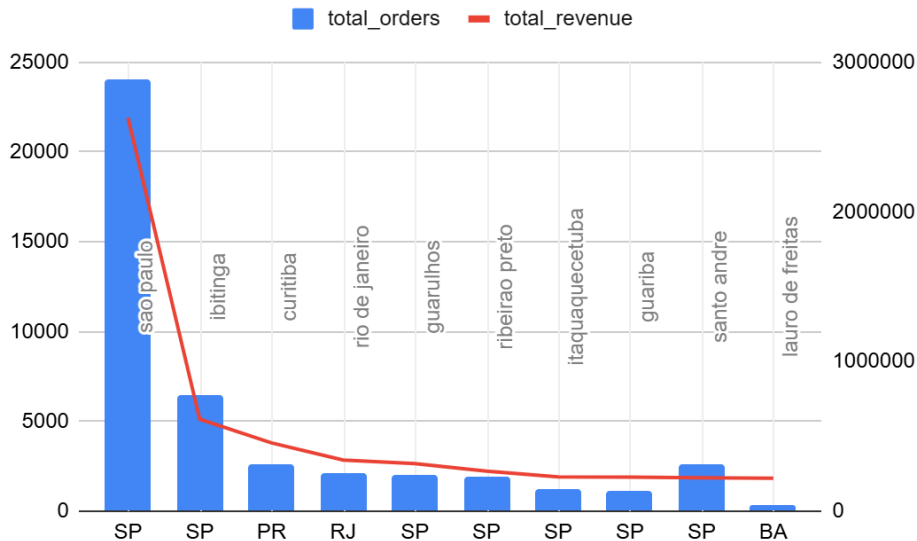
le wagon

Core Operational Risks

In the top 10 best performed sellers, 7 of 10 were in SP. This may have a potential influence in regional areas delivery.

Geographic Seller Concentration

Seller Performance: Top 10 Cities



le wagon

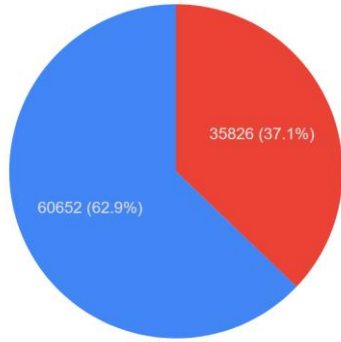
Results



le wagon

What makes the orders delayed?

Orders Exceeding Average Delivery Time



1,177 13,221,498.11

- Overall, 37.1% of the orders exceeded the average delivery time, whereas order approval is relatively fast.
- Over 60% of total seller revenue and order volume is concentrated in SP, creating regional logistics imbalances in regional area.

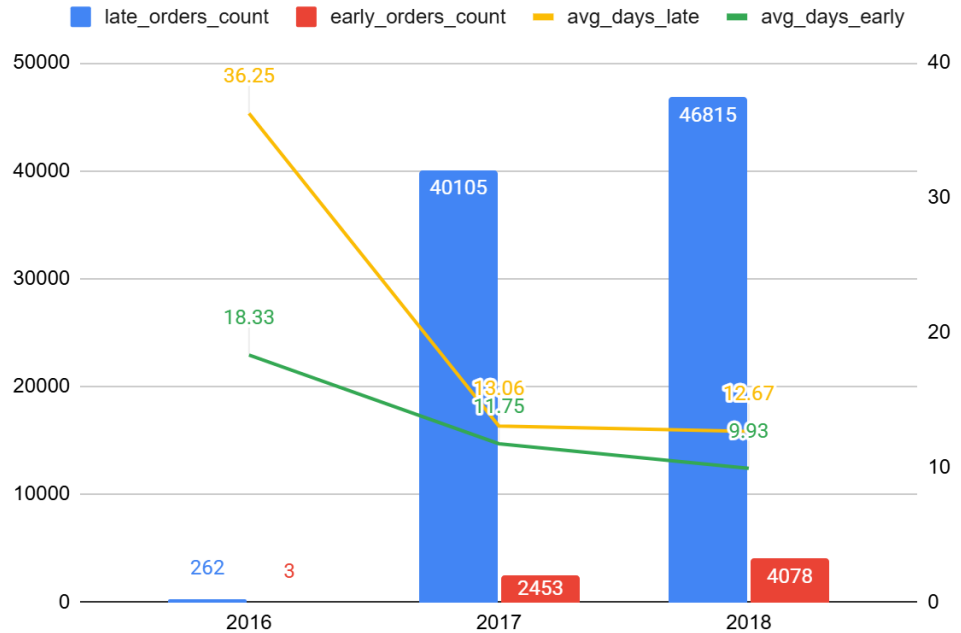


le wagon

What makes the orders delayed?

In 2017, over 40,000 orders were delayed while only 2,453 were delivered early and in 2018 these figures became 46,815 and 4,078, with average late delivery days around 13 days.

Order Delivery Performance & Volume (2016–2018)



le wagon

Recommendations



le wagon

Key Order Metrics & Recommendations

- Partner with sellers to synchronise warehouse processing times and make it available to the customers.
- Incentive seller acquisition in regional areas to decentralise fulfillment and lower cross-country shipping times.
- To increase the average items per order (currently 1.13) and AOV, automated recommendations at checkout can be used to encourage customers to add a second item to their basket.

Average Items per Order

1.13

Average Order Value (AOV)

135.54



le wagon

Next Steps

Next Steps & Further Analysis

- Product Category Impact: Analyse which specific product categories drive higher dispatch delays.
- Customer Review Correlation: Map delay days against review scores to quantify the correlation between delivery performance and ratings.



le wagon



le wagon